Supply Chain & E-commerce Analytics

Business Performance, Customer, Profitability & Delivery Analysis

PROJECT BUSINESS REPORT

Nitin

Tools: Python | Pandas | Jupyter Notebook | Power BI

Dataset: DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS

Project Type: Descriptive and Diagnostic Business Analytics

1. Executive Summary

SupplyChainIQ is a business analytics project built around the DataCo SMART Supply Chain dataset. The project was developed to understand how a large transactional dataset can be converted into useful business information. Rather than starting directly with Power BI visuals, I worked through the dataset in stages, starting with business understanding and data preparation, followed by feature engineering and exploratory data analysis in Python and Pandas.

The dataset contains 180,519 order-item records, representing approximately 65,752 distinct orders and 20,652 customers. Across the available historical period, the business records approximately $36.78 million in revenue and $3.97 million in profit. The average order value is approximately $559.45 and the overall late-delivery rate is approximately 57.33%.

The analysis highlights several areas that are more important than simply looking at total sales. Delivery reliability is a major operational concern, particularly for Second Class shipments. Profit margin also decreases as discount intensity increases in the dataset. Customer-frequency analysis shows that repeat and frequent customers contribute a much larger share of revenue than their share of the customer base. Category analysis further shows that high revenue does not automatically mean high profitability.

The final Power BI report was intentionally kept to four pages. Each page answers a different type of business question and avoids repeating the same visual analysis. The result is an interactive report that combines KPIs, trends, comparisons, rankings and contribution analysis.

> **Overall conclusion:** The strongest areas for management attention are delivery reliability, disciplined discounting, customer retention and evaluating category performance using both revenue and margin.

1.1 Project at a Glance

Item	Details
Dataset	DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS
Source	Kaggle
Records	180,519 order-item records
Distinct Orders	Approximately 65,752
Customers	Approximately 20,652
Revenue	Approximately $36.78M
Profit	Approximately $3.97M
Average Order Value	Approximately $559.45
Late Delivery Rate	Approximately 57.33%
Analytics Type	Descriptive and diagnostic analytics
Final Dashboard	4 Power BI pages

2. Business Problem & Objectives

2.1 Business Problem

Supply-chain businesses have to balance several outcomes at the same time. Revenue growth is important, but revenue alone does not show whether the business is creating strong economic value. A business can generate high sales through heavy discounting, low-margin products or customers who do not purchase repeatedly. Similarly, strong sales can be accompanied by poor delivery reliability, which can affect the customer experience and operational performance.

The project therefore treats the dataset as a business performance problem rather than only a visualization exercise. The analysis looks at financial performance, customer value, product and category performance, profitability drivers and delivery operations.

2.2 Core Business Areas

- Business performance: revenue, profit, profit margin, order volume and market contribution.
- Customer performance: customer segments, customer frequency, customer revenue and customer profitability.
- Product and category performance: revenue concentration, product contribution and category profitability.
- Commercial drivers: discount levels, order size, departments and market-level profitability.
- Delivery operations: shipping modes, delivery outcomes, late-delivery rates and average delivery delay.

2.3 Business Questions

1. How is the business performing overall in terms of revenue, profit and profit margin?
2. Which markets contribute most to revenue and how different are their profit margins?
3. Which customer segments generate the most revenue and profit?
4. Do repeat and frequent customers contribute a disproportionate share of revenue?
5. Which categories generate the most revenue and do they also generate healthy margins?
6. How is profit margin associated with different discount levels?
7. Which shipping modes have the most serious delivery issues?
8. Is late delivery concentrated in one market or spread across the business?
9. What business actions can reasonably be recommended from the observed patterns?

2.4 Analytical Approach

The project follows a staged workflow so that the dashboard is supported by analysis rather than being built in isolation.

Stage	**Purpose**
Business Understanding	Identify the business areas and questions that the dataset can support.
Data Preparation	Inspect, clean and prepare the source data.
Feature Engineering	Create analytical fields and useful business groupings.
EDA	Investigate distributions, trends, relationships and unusual patterns.
Power BI Modelling	Create the DateTable, relationship structure and measures.
Dashboard Development	Present the most useful findings through four focused pages.
Recommendations	Translate the observed patterns into practical business actions.

3. Dataset & Data Preparation

3.1 Dataset Overview

The project uses the DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS dataset downloaded from Kaggle. The dataset contains 180,519 records and 53 columns. The available fields cover orders, customers, products, markets, sales, profit, discounts, shipping modes and delivery information.

The source data is recorded at order-item level. This distinction is important because one order can contain multiple order items. For order-level metrics such as total orders and some delivery measures, the analysis therefore uses distinct Order IDs where appropriate rather than simply counting rows.

3.2 Data Organisation

The working project keeps the data in separate Raw, Cleaned and Engineered stages. The purpose of this structure is to make it clear how the original dataset moves through preparation before it is used for analysis.

Stage	Purpose
Raw	Original source data downloaded from Kaggle.
Cleaned	Prepared data after data-quality and formatting work.
Engineered	Analysis-ready data containing engineered fields used for EDA and Power BI.

3.3 Notebook Structure

Notebook	Main Purpose
01_Business_Understanding	Understand the dataset, business context and analytical questions.
02_Data_Preparation	Inspect and prepare the source data.
03_Feature_Engineering	Create fields required for deeper analysis.
04_EDA	Explore business performance, customers, products, profitability and delivery.

3.4 Data Quality and Analytical Checks

The analysis was not limited to calculating totals. I checked the structure of the data, examined the available fields and used exploratory analysis to identify patterns that could affect interpretation. In particular, the order-item grain was considered when calculating order-level metrics, and the sharp revenue decline near the end of the period was investigated rather than immediately being labelled as a business decline.

The project also includes a correction to the late-delivery calculation used during the analysis. The final dashboard uses the corrected late-delivery logic rather than the earlier incorrect calculation.

4. Exploratory Data Analysis

The EDA stage was carried out in Python using Pandas. The purpose was to understand the dataset before deciding which findings should become Power BI visuals.

4.1 Business Performance Analysis

The analysis examined revenue, profit, profit margin, orders, customers, average order value and quantity. Monthly trends were used to understand whether the business was stable over time or whether there were periods requiring further investigation.

4.2 Revenue Trend

Revenue remained around the $1 million monthly level for much of the available period. However, a sharp decline appears toward the end of the dataset: approximately $1.074M in October 2017, $0.627M in November, $0.504M in December and $0.332M in January 2018.

> **Interpretation:** This decline should not automatically be treated as a genuine decline in demand. Order coverage and volume also change substantially during the same period, so the final months should be validated before using the trend for a management decision.

4.3 Market Analysis

Market	Revenue Contribution
Europe	29.56%
LATAM	27.94%
Pacific Asia	22.49%
USCA	13.77%
Africa	6.24%

Europe and LATAM together contribute approximately 57.50% of total revenue. Adding Pacific Asia brings the combined contribution to approximately 80%. Despite these differences in revenue scale, market-level profit margins are relatively close, suggesting that the main difference between markets is their business scale rather than a large profitability gap.

4.4 Customer Segment Analysis

Segment	Revenue	Profit
Consumer	~$17.16M	~$2.07M
Corporate	~$10.03M	~$1.20M
Home Office	~$5.86M	~$0.69M

Consumer is the largest segment by both revenue and profit. Corporate is the second-largest segment, followed by Home Office. This makes customer segmentation useful for understanding where the business is generating its commercial value.

5. Customer, Product & Category Findings

5.1 Customer Frequency

Customer Group	Customer Share	Revenue Share
One-time	43.02%	7.60%
Occasional	16.23%	14.15%
Repeat	29.85%	48.87%
Frequent	10.91%	29.38%

Repeat and frequent customers represent approximately 40.76% of customers but generate approximately 78.25% of revenue. The difference between customer share and revenue share is one of the strongest commercial findings in the analysis.

At the individual-customer level, however, the business is not heavily dependent on a small handful of customers. The top 10 customers contribute only approximately 0.25% of total revenue, and the top individual customer contributes approximately 0.0286%. The important concentration is therefore in purchasing frequency, not in a few named customers.

5.2 Product and Category Performance

The dataset contains 118 products across 50 categories. The highest-revenue categories include:

Category	Approx. Revenue
Fishing	$6.93M
Cleats	$4.43M
Camping & Hiking	$4.12M
Cardio Equipment	$3.69M
Women's Apparel	$3.15M
Water Sports	$3.11M

High revenue does not always correspond to the strongest margin. Several high-revenue categories, including Camping & Hiking, Cardio Equipment, Water Sports, Shop By Sport and Computers, show profit margins below the overall business margin. This is why the final dashboard compares revenue with profit margin rather than ranking categories only by sales.

5.3 Unique Products Over Time

Unique-product analysis was included to understand how the number of distinct products represented in the order data changes over time. This analysis becomes particularly relevant near the end of the dataset because product coverage and order composition change alongside the fall in revenue.

6. Profitability & Commercial Drivers

6.1 Profitability and Discounting

Discount Category	Profit Margin
No Discount	13.09%
Low	11.48%
Medium	10.16%
High	9.03%

Profit margin decreases consistently as discount intensity increases. The difference between no discount and high discount is approximately 4 percentage points. The pattern is commercially important because additional sales generated through discounting should be evaluated against the margin sacrificed.

> **Important limitation:** The relationship is observational. It shows an association between discount category and profit margin, but it does not prove that discounts alone caused the lower margin. Product mix, pricing, costs and order characteristics can also influence the result.

6.2 Order Size and Profit Contribution

The dashboard also examines profit contribution by order size. The purpose is to compare the economic contribution of Small, Medium and Large orders rather than looking only at order counts.

6.3 Department-Level Profit

Department-level analysis shows strong differences in total profit contribution. Fan Shop contributes the largest share, followed by Apparel, Golf and Footwear. Smaller departments contribute comparatively little to total profit and can therefore be prioritised differently from the major profit-generating departments.

6.4 Monthly Profit Margin

Monthly profit margin varies over time rather than remaining constant. The dashboard includes the monthly trend to help identify periods of stronger and weaker profitability without confusing the margin trend with the much larger revenue scale.

7. Delivery & Operations Analysis

7.1 Overall Delivery Outcomes

Outcome	Approx. Records
Late	103,400
Early	43,366
On Time	33,753

Approximately 57.33% of orders are classified as late. This makes delivery reliability the strongest operational concern identified in the project.

7.2 Shipping Mode Performance

Shipping Mode	Late Delivery Rate	Average Delay
First Class	100%	~1 day
Second Class	~80%	~2 days
Same Day	~48%	~0.5 day
Standard Class	~40%	~0 days

Second Class is particularly important because it combines a high late-delivery rate with the highest average delay among the major shipping modes. First Class shows a 100% late rate in the available records, but this should be treated as a dataset-specific observation rather than a general statement about First Class shipping.

7.3 Market-Level Delivery

Late-delivery rates are close across markets, ranging from approximately 56.6% to 57.6%. The small difference between markets suggests that the delivery problem is not primarily geographic. The larger variation appears across shipping modes, which makes shipping and fulfilment processes a more useful first area of investigation.

7.4 Average Delivery Delay

Average delay also varies by shipping mode. Second Class has the highest average delay at about two days, followed by First Class at about one day and Same Day at about half a day. Standard Class is close to zero in the observed data.

8. Power BI Dashboard

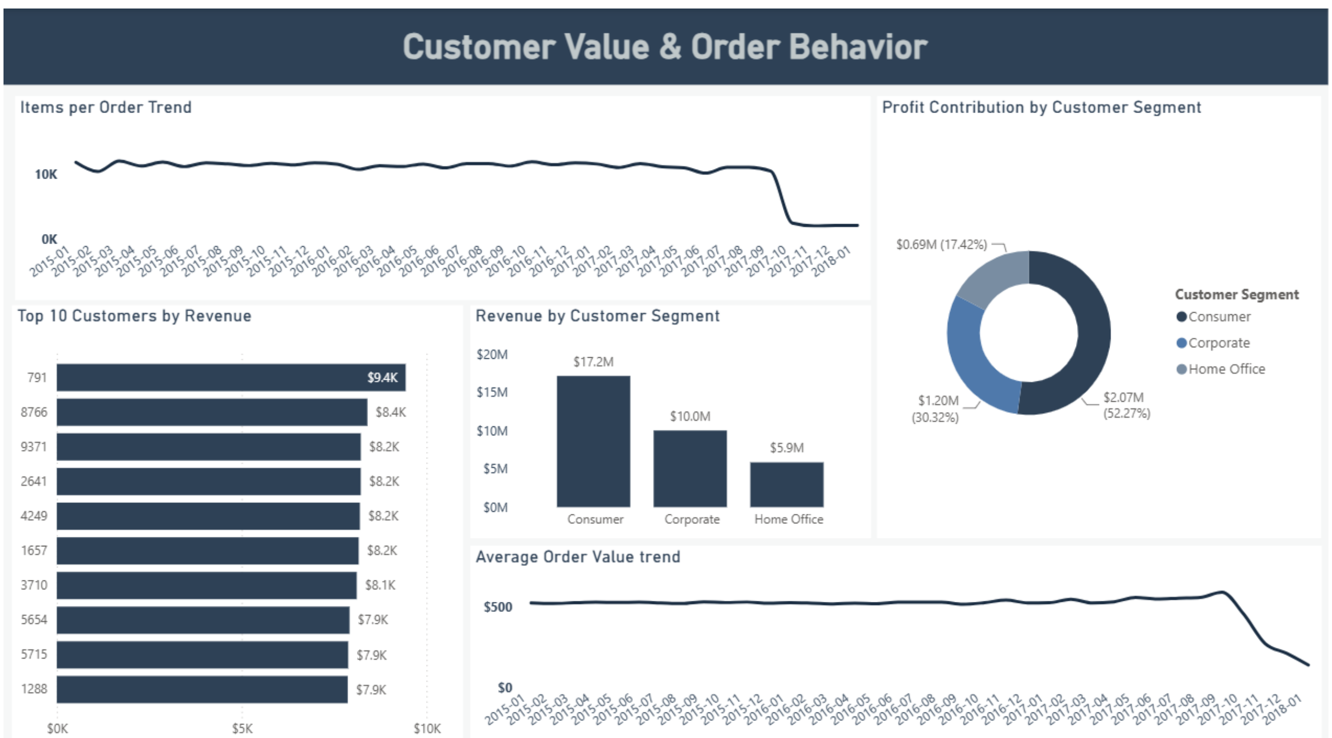
The final Power BI dashboard contains four pages. The pages were designed separately so that each one focuses on a distinct business area and the report does not repeatedly show the same type of chart.

8.1 Supply Chain Executive Overview

- The **Supply Chain Executive Overview** is the main landing page of the Power BI report. It is designed to give management a quick understanding of the overall health of the business before moving into the more detailed customer, commercial and operational analysis presented on the following pages.
- The page combines high-level KPIs with market-level comparisons, a monthly revenue trend and an overview of delivery performance. The intention is not to provide every available metric on one page, but to highlight the indicators that are most useful for understanding the current business position.
    - Total Revenue
    - Total Orders
    - Total Profit
    - Total Customers
    - Average Order Value
    - Late Delivery Rate
    - Monthly Revenue Trend
    - Profit Margin by Market
    - Revenue by Market
    - Delivery Performance

8.2 Customer Value & Order Behavior

- The **Customer Value & Order Behavior** page focuses on understanding how different customer groups contribute to the business and how their purchasing behaviour changes over time. It goes beyond overall revenue by looking at customer segments, individual customer contribution and order-level behaviour.
- The page combines customer-segment comparisons, customer-level ranking and monthly trends to provide a clearer picture of customer value and purchasing patterns.
    - **Revenue by Customer Segment** – compares the revenue generated by Consumer, Corporate and Home Office customers and identifies the segment contributing the most revenue.
    - **Customer Profit Mix** – shows how profit is distributed across the different customer segments, allowing revenue contribution to be compared with actual profitability.
    - **Top 10 Customers by Revenue** – identifies the highest-revenue individual customers and helps understand whether revenue is concentrated among a small number of customers.
    - **Average Order Value Trend** – tracks changes in the average value of an order over time and helps identify changes in purchasing value.
    - **Items per Order Trend** – shows how the average number of items purchased per order changes over time and provides another view of customer purchasing behaviour.
- Overall, this page helps answer **which customers are contributing the most value and how their purchasing behaviour changes over time**, while keeping the analysis separate from the broader business overview

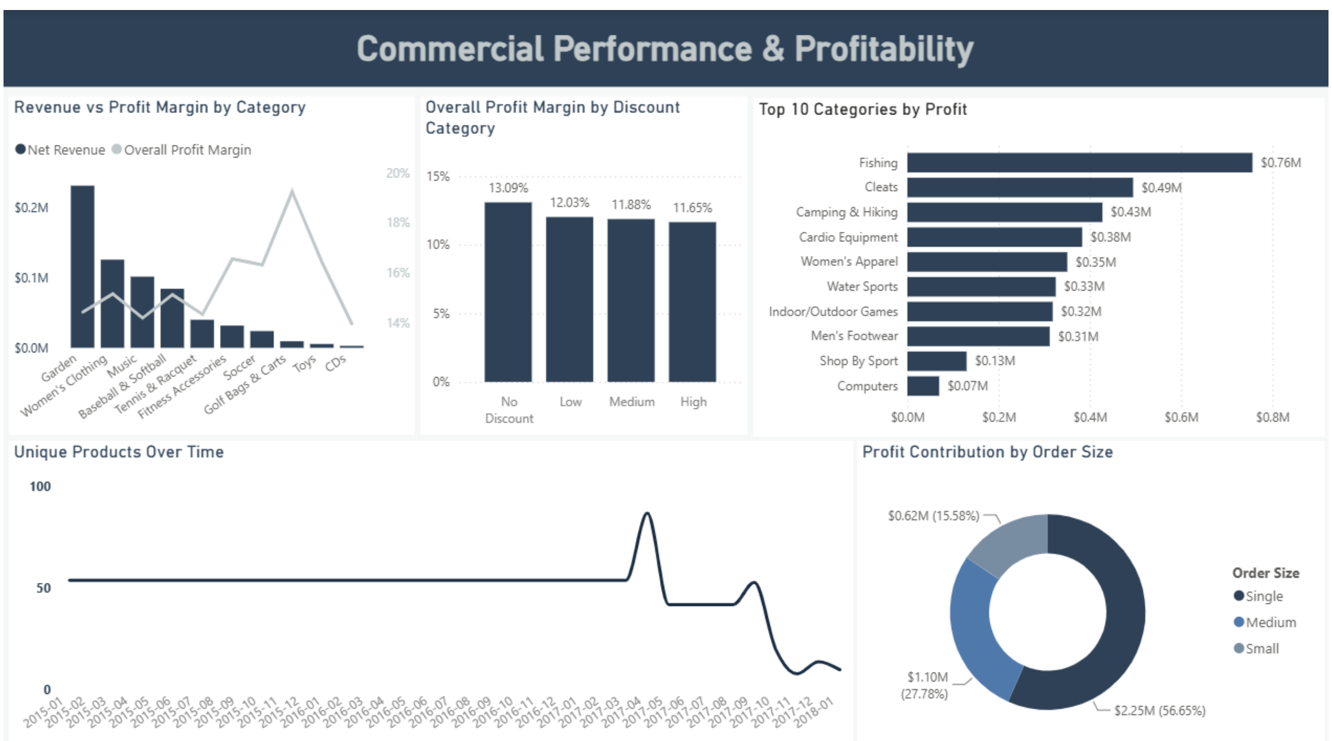


8.3 Commercial Performance & Profitability

- The **Commercial Performance & Profitability** page focuses on understanding how different commercial factors affect revenue and profitability. It brings together category performance, discounting, order size and product availability to identify where the business is generating value and where profitability may be under pressure.

  - **Revenue vs Profit Margin by Category** – compares category revenue with profit margin to identify categories that generate high sales but may have relatively lower profitability.
  - **Profit Margin by Discount Level** – shows how profit margin changes across different discount levels and helps evaluate the relationship between discounting and profitability.
  - **Profit Contribution by Order Size** – compares the profit generated by different order-size groups and shows which type of orders contributes more to overall profit.
  - **Unique Products Over Time** – tracks the number of distinct products appearing in orders over time and helps identify changes in product coverage and order composition.
  - **Top 10 Categories by Profit** – ranks categories based on total profit and highlights the categories contributing the most to the business's profitability.

- Overall, this page helps answer **where revenue and profit are being generated and which commercial factors are influencing profitability**, while providing a more detailed view of the business than the executive overview.

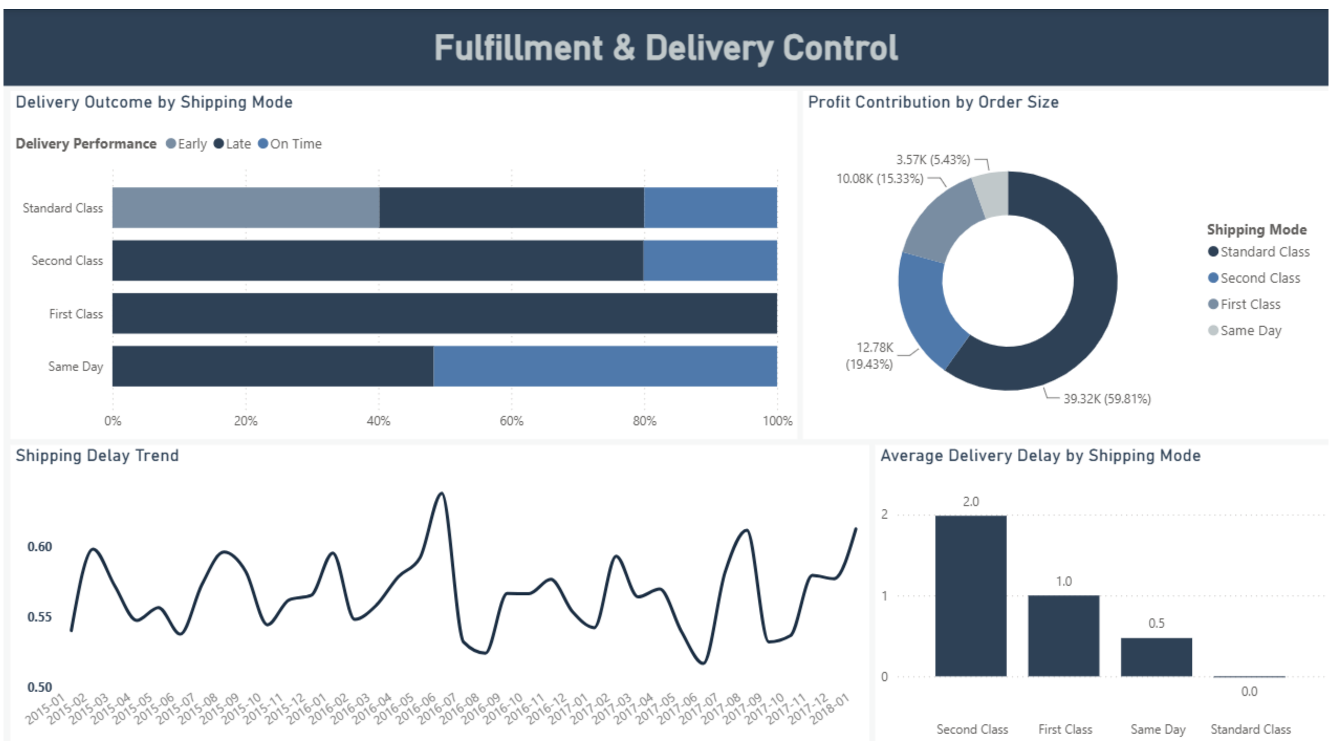


8.4 Fulfillment & Delivery Control

- The fourth page focuses on **operational execution and delivery performance**. It analyses shipping modes, delivery outcomes and delay patterns to identify where fulfilment performance may require attention.

  - **Delivery Outcome by Shipping Mode** – compares late, early and on-time deliveries across different shipping modes and highlights which modes have weaker delivery performance.
  - **Average Delivery Delay by Shipping Mode** – compares the average delay associated with each shipping mode and helps identify where delays are more severe.
  - **Order Mix by Shipping Mode** – shows how orders are distributed across the available shipping modes and provides context for the delivery-performance results.
  - **Shipping Delay Trend** – tracks delivery delays over time and helps identify whether operational performance is improving, worsening or remaining relatively stable.

- Overall, this page helps answer **which shipping modes are contributing to delivery issues and how fulfilment performance changes over time**, giving a more detailed operational view than the delivery KPI shown on the Executive Overview page.



9. Power BI Data Model & DAX

9.1 DateTable

A dedicated **DateTable** was created for time-based analysis. It contains **Date, Day, Month, Month No., Quarter, Week No. and Year**.

The Month No. column is used to sort the month names in chronological order instead of alphabetical order. This is important for visuals such as monthly revenue, average order value and profit-margin trends, where the months need to appear in their natural sequence.

9.2 Relationship

The DateTable is connected to the engineered dataset through the order date using an active one-to-many relationship.

From	Cardinality	To
DateTable[Date]	1 : *	Featured_Engineering[Order Date]

This relationship allows the DateTable to act as the main time dimension for the report. It is used by the monthly trend visuals and ensures that date-based calculations respond correctly to filters and slicers.

9.3 Measures

Several DAX measures were created to calculate the main KPIs and analytical metrics used throughout the dashboard.

- **Net Revenue** – calculates the total revenue used for the main business performance analysis.
- **Total Profit** – measures the overall profit generated from the available orders.
- **Total Orders** – calculates the number of distinct orders.
- **Total Customers** – calculates the number of distinct customers.
- **Average Order Value** – measures the average revenue generated per order.
- **Overall Profit Margin** – calculates profit as a percentage of revenue.
- **Total Quantity Sold** – measures the total quantity of items sold.
- **Average Delivery Delay** – calculates the average delivery delay.
- **Late Delivery Rate** – measures the proportion of orders classified as late.

9.4 Dashboard Design Decisions

Visual selection was based on the type and amount of information being compared. **Bar charts** were mainly used for category comparisons and rankings, while **line charts** were used for trends over time. **Donut charts** were used only when the number of categories was small enough to remain clear and readable.

I also avoided placing measures with very different scales on the same axis when doing so would make one of the measures difficult to interpret. For example, revenue and profit can have a large difference in magnitude, so combining them without considering their scales can make the smaller metric appear almost flat.

The dashboard was also limited to **four pages**. Instead of continuously adding pages and repeating similar visuals, each page was given a specific purpose: executive performance, customer behaviour, commercial profitability, and fulfillment and delivery. This keeps the report more focused and easier to navigate.

10. Business Recommendations

10.1 Prioritize Delivery Reliability for Second Class Shipping

Evidence: Second Class has an approximately 80% late-delivery rate and an average delay of about two days.

Recommendation: Investigate carrier performance, fulfilment processing time, scheduling assumptions and transportation capacity. Service-level targets should also be reviewed.

Expected impact: Better reliability could reduce late deliveries, improve customer experience and reduce operational disruption.

10.2 Review Discounting Based on Profitability

Evidence: Profit margin falls from 13.09% without discounts to 9.03% for high discounts.

Recommendation: Evaluate promotions using incremental profit rather than revenue alone. High discounts should be used where additional volume justifies the margin reduction.

Expected impact: A more disciplined discounting strategy can help distinguish productive promotions from promotions that mainly reduce margin.

10.3 Focus Retention on Repeat and Frequent Customers

Evidence: Repeat and frequent customers represent about 40.76% of customers but generate about 78.25% of revenue.

Recommendation: Focus retention efforts on converting occasional and one-time customers into repeat purchasers while maintaining frequent customers.

Expected impact: Increasing repeat purchase frequency can increase revenue without relying entirely on acquiring new customers.

10.4 Evaluate Categories Using Revenue and Margin

Evidence: Several high-revenue categories have margins below the overall business margin.

Recommendation: Compare revenue, profit and margin together and investigate pricing, discounting, product costs and product mix in high-revenue but lower-margin categories.

Expected impact: This helps management distinguish high-volume categories from categories that create stronger economic value.

10.5 Validate the Late-Period Revenue Decline

Evidence: Revenue falls sharply from October 2017 to January 2018 while order coverage also changes.

Recommendation: Validate completeness and consistency of the final months before treating the decline as a genuine demand or business-performance problem.

Expected impact: This reduces the risk of making a decision from incomplete or structurally different data.

11. Data Limitations

11.1 Historical Dataset

The analysis uses a historical DataCo dataset rather than live operational data. The findings describe patterns in the available period and should not automatically be treated as current business conditions.

11.2 Late-Period Revenue Coverage

Revenue falls sharply near the end of the dataset while order coverage also changes. The final months should therefore be validated before the decline is used as evidence of a real demand problem.

11.3 Observational Relationships

The analysis identifies associations rather than proving causality. For example, higher discount categories are associated with lower profit margins, but other variables may contribute to that relationship.

11.4 Order-Item Level Data

The source dataset contains order-item records rather than one row per order. Order-level and customer-level metrics must therefore account for the underlying grain and use distinct identifiers where appropriate.

11.5 Delivery Metrics

Delivery results can differ depending on whether calculations are performed at order-item level or distinct-order level. The dashboard uses distinct orders where appropriate for order-level metrics.

11.6 No Predictive Modeling

This project focuses on descriptive and diagnostic analytics. It identifies historical patterns and potential business actions but does not predict future sales, delivery risk or customer behaviour.

12. Conclusion

SupplyChainIQ provides a business-focused view of the DataCo SMART Supply Chain dataset across financial performance, customers, products, profitability and delivery operations. The project moves from raw transactional data to a structured analytical output using Python, Pandas, feature engineering, exploratory analysis and Power BI.

The analysis shows that the business generates substantial revenue and profit, but delivery reliability is the clearest operational weakness. The high late-delivery rate is visible across markets, while shipping-mode differences are much larger. This makes shipping and fulfilment processes a more useful starting point for operational investigation.

Commercially, the discount analysis shows a consistent association between higher discount levels and lower profit margins. Customer analysis also shows that repeat and frequent customers generate a disproportionately large share of revenue. At the same time, the top individual customers do not account for a large enough share of revenue to indicate dependence on a small number of customers.

The category analysis reinforces the importance of evaluating revenue and profitability together. Some categories generate strong sales but lower margins, so prioritising categories purely by revenue could lead to incomplete business decisions.

Overall, the project demonstrates a complete descriptive analytics workflow: understanding the business problem, preparing the data, engineering useful features, exploring the dataset, building a Power BI model and DAX measures, creating an interactive four-page dashboard, and translating the results into practical recommendations.

13. Project Structure & Files

The final project is organised so that the working analysis, dashboard, documentation and screenshots are kept separate.

Folder / File	Contents
data/	Raw, cleaned and engineered datasets used locally during the project.
notebooks/	01 Business Understanding, 02 Data Preparation, 03 Feature Engineering and 04 EDA.
powerbi dashboard/	Final Power BI dashboard file.
reports/	Final project report.
screenshots/	Final screenshots of the four dashboard pages.
README.md	GitHub project overview, workflow, findings and setup information.

13.1 GitHub Repository

The GitHub repository contains the project documentation, notebooks, Power BI dashboard, report and final screenshots. The large source data files are kept outside the GitHub repository because the browser upload limit makes the large files unsuitable for a normal repository upload, and the original dataset can be obtained from its Kaggle source.

13.2 Future Enhancement: Dynamic Analysis

A possible future improvement is a Dynamic Analysis page using Power BI Field Parameters. The user could choose a dimension such as Category, Market, Customer Segment or Shipping Mode and then choose a measure such as Revenue, Profit, Orders, Quantity or Profit Margin. This would reduce the need to create separate fixed visuals for every possible combination.